Helix Forge Research Failure Analysis Research Brief

HELIX FORGE LAB / p04-full-004538 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded failure analysis evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000055 (research/papers/2026-07-13/helix-forge-research/failure-analysis/tables/helix-forge-research-failure-analysis-thresholds-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	184	0.886	0.863	0.874
slice-02	161	0.868	0.876	0.872
slice-03	168	0.885	0.852	0.868
slice-04	238	0.867	0.865	0.866
slice-05	244	0.884	0.878	0.881